

8a	<u>Fast moving/ high speed/ high (kinetic) energy</u> electrons moving towards the target metal (anode), undergo <u>rapid/large deceleration/ acceleration</u> as they interact with the lattice ions. The loss in energy of the electrons is emitted in the form of X-ray photons, and since <u>the loss of energy can vary according to the interaction</u>, they form the continuous X-ray spectrum.	B1 M1 A1
8b	tungsten is ductile so it can be made into thin wires tungsten has a high atomic number tungsten is a good thermionic emitter / good at emitting electrons tungsten has a very high melting point (any one relevant point)	A1
8ci1.	From Fig. 8.2, μ of muscle is 0.40 cm^{-1}. Using $\frac{I}{I_0} = e^{-\mu x}$, ratio along MM = $e^{-0.40(8.0)}$ = 0.041 or 0.0408	C1 A1
8ci2.	From Fig. 8.2, μ of bone is 2.46 cm^{-1}. Using $\frac{I}{I_0} = e^{-\mu x}$,	

	ratio along BB = $\left(\frac{I}{I_0}\right)_{\text{muscle}} \times \left(\frac{I}{I_0}\right)_{\text{bone}}$ $= e^{-0.40(4.0)} \times e^{-2.46(4.0)}$ $= 0.000011 \text{ or } 0.0000108$ or ratio along BB = $\left(\frac{I}{I_0}\right)_{\text{muscle, 2 cm}} \times \left(\frac{I}{I_0}\right)_{\text{bone}} \times \left(\frac{I}{I_0}\right)_{\text{muscle, 2 cm}}$ $= e^{-0.40(2.0)} \times e^{-2.46(4.0)} \times e^{-0.40(2.0)}$ $= 0.000011 \text{ or } 0.0000108$	C1 C1 A1
8cii	Since the ratio along BB is <u>3 orders of magnitude smaller</u> than that along MM, the contrast is <u>good</u> .	M1 A1
8d	MRI is able to provide excellent contrast between the different parts of the brain (gray matter, white matter and blood) due to the distinct relaxation times in Fig. 8.5, whereas X-ray has poor contrast between blood and brain due to similar absorption coefficients. MRI is safer as it uses magnetic field and radio frequency waves unlike X-rays which is an ionising radiation and can cause tissue damage causing mutations which can lead to growth of cancerous tissues.	B1 B1
8e	• small space, not suitable for patient who are claustrophobic • strong magnetic field can cause ferromagnetic objects in the room to become projectile posing risk to patient and staff • metal implants (e.g. pacemaker, cochlear implants) / ferromagnetic implants in patients may malfunction / move in the strong magnetic field • (long scan time) patient needs to remain still for long duration and any reasonable challenges e.g. any movement may cause blurred image, children or patients who cannot lie down for long duration. • ineffective for bone imaging due to fast relaxation times • helium gas leaking during a quench resulting in suffocation or frostbite (any one relevant point)	A1
8f	For a solenoid, $B = \mu_0 n I$ $1.5 = (4\pi \times 10^{-7}) \times n \times 750$ $n = 1591.55 \text{ m}^{-1}$ Hence, number of turns = $n \times 2.2$ $= 3500 \text{ turns}$	C1 A1
8g	Lamor frequency, $\omega = \gamma B_0$ $= (2.68 \times 10^8) \times 1.5 = 4.02 \times 10^8 \text{ rad s}^{-1}$ frequency of RF wave = $\omega/2\pi$ $= 4.02 \times 10^8 / 2\pi$ $= 64 \text{ or } 64.0 \text{ MHz}$	C1 A1
8h	assuming all electrical energy is used to vaporise liquid helium, rate of electrical heating = rate of vaporisation of liquid helium $I^2 R = m l_v$ where m is the initial rate of vaporization $(750)^2 (0.0045) = m (21 \times 10^3)$ $m = 0.12 \text{ or } 0.121 \text{ kg s}^{-1}$	C1 A1

